

Movie/Series Search and Retrieval Agent

Project Overview

This project enables the retrieval of movie and TV show metadata based on a search query. The metadata is stored in a Pinecone vector database, which is used for efficient, semantic search over movie/TV show titles. If the metadata is not found in Pinecone, the system queries the OMDb API to fetch the details and stores them in Pinecone for future retrieval. The metadata includes information such as the title, year, genre, director, cast, IMDB rating, and a trailer link.

Setup Instructions

Prerequisites

- Python 3.8 or higher
- API Keys for OpenAI, OMDb, and Pinecone
- Install required Python packages using `pip` from the `requirements.txt` file

Steps for Setup

1. **Clone the repository:**

```
git clone https://github.com/OyeBI-01/RAG-Agent-for-Movie-and-Series-Search.git
cd RAG-Agent-for-Movie-and-Series-Search
```

2. **Create a `.env` file** in the root directory of the project. It should contain the following keys:

```
OPENAI_API_KEY=your_openai_api_key
OMDB_API_KEY=your_omdb_api_key
PINECONE_API_KEY=your_pinecone_api_key
INDEX_NAME=your_pinecone_index_name
```

3. **Install the dependencies:** Run the following command to install the necessary libraries:

```
pip install -r requirements.txt
```

4. Initialize Pinecone:

- The script will automatically initialize Pinecone using your provided API key and index name. It will check whether the specified index exists and has the correct dimension. If the index does not exist or the dimensions are incorrect, the script will create or delete the index accordingly.

5. Run the Application: To test the system, run the following command:

```
python app.py
```

- You will be prompted to enter whether you're searching for a movie or TV series, followed by the title of the movie/TV show.
- The system will return metadata for the title, either retrieved from Pinecone or fetched from the OMDb API.

Example Input/Output

Input:

Enter 'movie' or 'series': movie
Enter the title: Interstellar

Output:

Agent Response:

```
{  
  "Title": "Interstellar",  
  "Year": "2014",  
  "Director": "Christopher Nolan",  
  "Cast": "Matthew McConaughey, Anne Hathaway, Jessica Chastain",  
  "IMDB Rating": "8.6",  
  "Genre": "Adventure, Drama, Sci-Fi",  
  "Plot": "When Earth becomes uninhabitable, an ex-NASA pilot leads a mission to find a new home for  
humanity.",  
  "Trailer": "https://www.youtube.com/results?search_query=Interstellar+trailer"  
}
```

Challenges Faced

1. **Pinecone Index Management:**

- Initially, I faced challenges in ensuring that the Pinecone index had the correct dimensions for storing and querying the embeddings. The Pinecone API required a specific dimension (1536) for the embeddings generated by OpenAI, so I had to ensure that the index either already existed with the correct dimensions or was deleted and recreated with the proper settings.

2. **Handling API Errors:**

- While integrating the OMDb API, I encountered issues with rate limiting and occasional API downtimes. I had to implement proper error handling to ensure that the system gracefully handles any failures when querying OMDb and falls back to Pinecone when possible.

3. **Embedding Consistency:**

- Ensuring consistent embedding generation was tricky, especially when dealing with queries that had slight variations in wording. To address this, I utilized a cosine similarity metric in Pinecone to find the best matches and adjusted the threshold for matching metadata from Pinecone to avoid mismatches.

4. **Storing Metadata in Pinecone:**

- Storing metadata in Pinecone, especially with a unique document ID for each title, required careful handling to ensure that entries weren't duplicated or overwritten by future data. I implemented a check based on the title and media type to ensure that only unique entries were stored and retrieved.

Notes

- **Pinecone:** This is used as the vector database for storing and retrieving the movie/TV show metadata. It allows for fast and efficient semantic search.
- **OMDb API:** Used to fetch movie/TV show metadata when it's not found in Pinecone.
- **OpenAI API:** Used for generating embeddings of the titles for searching similar documents in Pinecone.